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Team Details

Team Identification

Team Name:
Zahrathustra

Team Leader:
Gantan Etika Murty

Members Qualification:
Working Professionals & Founders (21+)

Problem Statement

“Over 64 million Indonesian MSMEs lack affordable back-office staff. Hiring an accountant costs Rp 4M-8M/month, while manual spreadsheets cause statutory tax fines (PP 55/2022) and unbankable financial statements.”

Nata delivers an autonomous 4-bot digital staff on Google Cloud that automates bookkeeping, 0.5% MSME tax defense, and verified licensing while owners sleep.

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Brief About the Idea

Opportunities & Problem

- **64M MSMEs** generate 61% of Indonesian GDP, yet 94% lack formal accounting.
- E-commerce sellers struggle with escrow deductions and **PMK 37/2025** withholding tax.
- **Unit Economics:** Small merchants cannot afford Rp 4M-8M/month accounting retainers.

Unique Selling Proposition (USP)

Sell convenience first, statutory compliance second.
Zero bureaucratic jargon on top (Canva-like ease); mathematical SAK EMKM zero-delta equilibrium and verbatim statute citations under the hood.

How It Solves the Problem: The 4-Bot Staff

1. **Akuntan Pribadi (Agent Keuangan):** Multimodal OCR parsing to balanced SAK EMKM ledgers.
2. **Konsultan Pajak (Agent Fiskal):** Shields Rp 500M exemption; computes 0.5% PPh Final.
3. **Spesialis Legal (Agent Regulasi):** Verifies operations against official 1,790 KBLI 2020 codes.
4. **Manajer Keuangan (Orchestrator):** Coordinates atomic handoffs and 24/7 background audit routines.

Key Differentiation: Deterministic Guardrails

Unlike raw LLM wrappers that hallucinate, **the model proposes, deterministic Python enforces.** Math delta must equal 0 before any journal entry is saved.

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List of Features Offered by the Solution

Receipt to Ledger

Instant multimodal image extraction of raw receipts into balanced double-entry SAK EMKM journal entries.

PP 55 Tax Sentinel

Monitors turnover against the Rp 500M exemption and computes 0.5% PPh Final with exact legal citations.

Scheduled Routines

24/7 background agent jobs for morning CFO cashflow briefings and marketplace reconciliation.

Bank-Ready Reports

One-click generation of balance sheets, income statements, and notes exportable for bank loans (KUR).

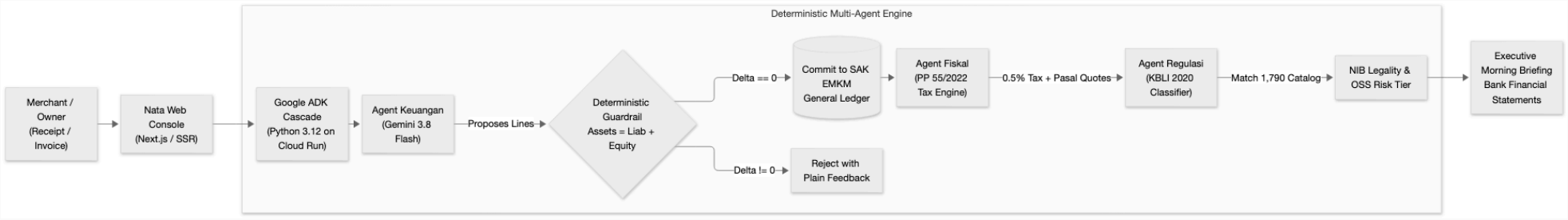
KBLI 2020 Classifier

Maps natural business descriptions to verified 5-digit KBLI codes and official OSS risk levels.

Provenance & i18n

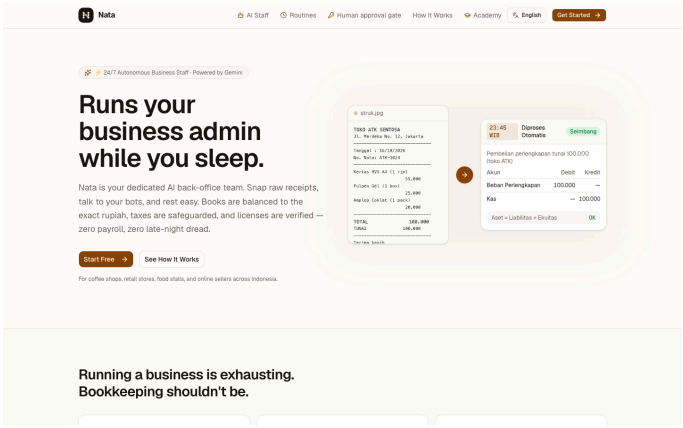
3-tier proof chain (“Kenapa Angka Ini?”) with 1-click resolution; bilingual EN/ID switcher.

Process Flow Diagram / Use-Case Flow



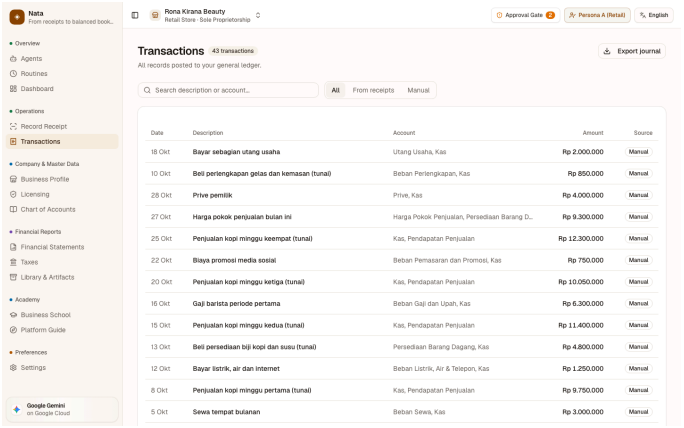
User Experience & Live Surface Design

Landing Page: Transformation Proof



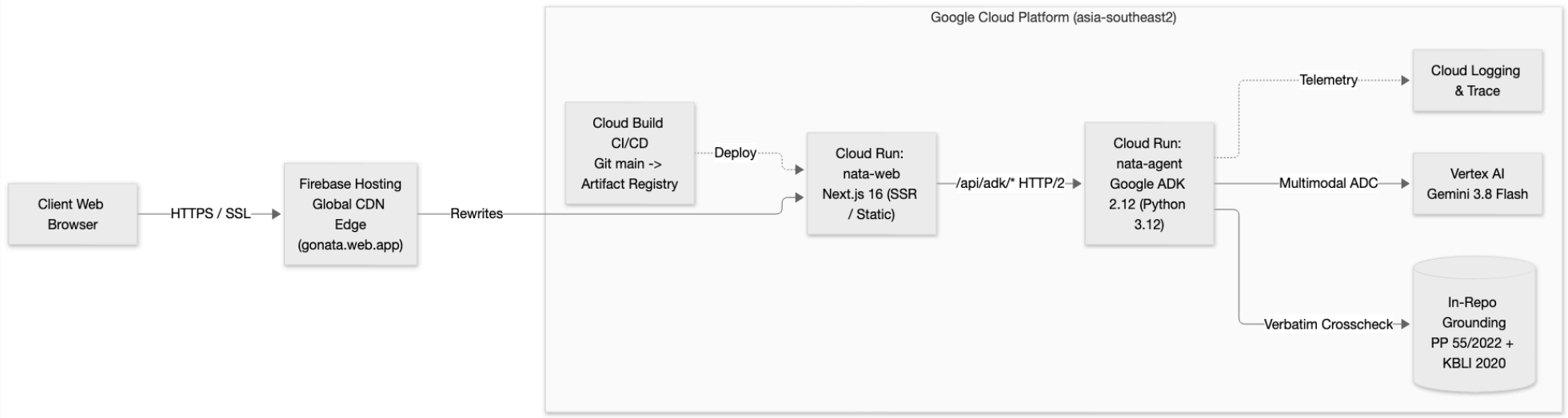
Visual proof: Crumpled receipt side-by-side with balanced journal entry.

Interactive Live Ledger & Transactions



Zero-delta double entry commitment with SAK EMKM account classification.

Architecture Diagram of the Proposed Solution



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Technologies Used in the Solution

Google Cloud Platform (Native Stack)

- **Google Agent Development Kit (ADK):** Python multi-agent cascade orchestration, session handling, and schema validation.
- **Gemini 3.8 Flash via Vertex AI:** High-throughput multimodal parsing and reasoning authenticated via ADC.
- **Google Cloud Run (Serverless):** Next.js web console and Python ADK agents with automated autoscaling.
- **Firebase Hosting:** Global CDN edge distributing assets, rewrites, and SSL termination.
- **Cloud Build:** Automated CI/CD from git main to Artifact Registry containers.
- **Google Cloud Journey TTS:** Studio-grade voice synthesis (`en-US-Journey-F`) for video walkthroughs.

Frontend & Deterministic Foundation

- **Next.js 16 (React 19) & Tailwind CSS:** Solid opaque surfaces (zero glassmorphism), strict accessibility standards, and sub-12px typography ban.
- **Python 3.12 (uv):** Deterministic SAK EMKM financial calculations and in-repo statutory cross-checking.
- **Playwright E2E Suite:** Automated UI testing, live crawl verification, and synchronized demo capture.

Estimated Implementation Cost

Serverless Cloud Unit Economics (1,000 Active MSMEs)

Component	Usage Profile	Monthly Cost
Cloud Run (Web & Agent)	Scale-to-zero compute	Rp 120.000
Vertex AI (Gemini 2.5)	300 receipts/biz/mo	Rp 95.000
Firebase CDN & Artifacts	Global edge caching	Rp 35.000
Total Operational Cost	**1,000 Businesses**	**Rp 250.000**

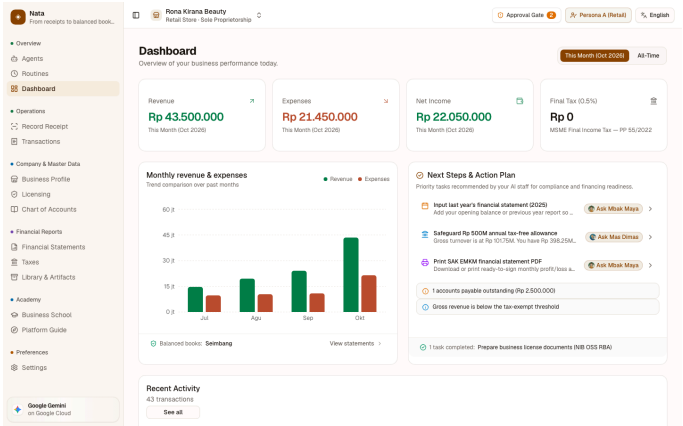
Infrastructure Cost Per Merchant: Rp 250 / month ($0.016 USD)

Merchant Economic Value

- **Traditional Cost:** Rp 4.000.000 - Rp 8.000.000/month for a dedicated accountant.
- **Nata Pro Plan:** Rp 99.000/month.
- **Merchant ROI:** Over **97% immediate cost reduction** with bank-ready compliance.

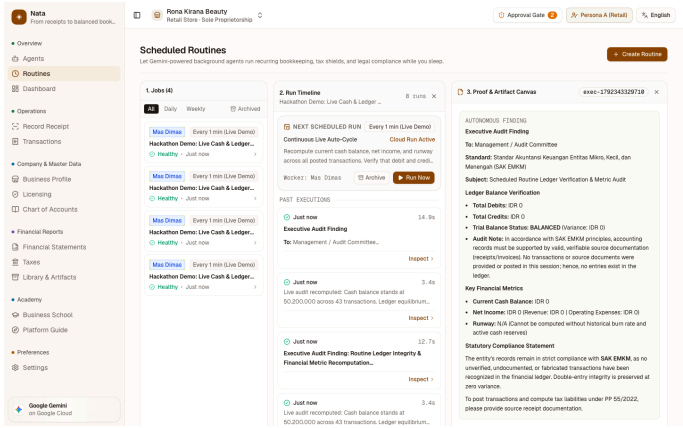
Snapshots of the Live Prototype

Executive Dasbor & Morning Briefing



CFO morning greeting, live cash runway, and 1-click deterministic actionables.

Autonomous Scheduled Routines Engine



3-pane background worker executing 1-minute live jury demo recomputation.

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Prototype Performance & Quality Benchmarks

Automated Test Suite (100% Pass Rate)

- **Python ADK & Core Engine:** 156/156 pytest unit and integration tests passing cleanly (10.8s runtime).
- **Web & Entity Suite:** 68/68 node/typescript verification assertions passing (399ms runtime).
- **TypeScript & ESLint:** `tsc --noEmit` and `eslint` clean with zero warnings.
- **Live Production Crawl:** 16/16 routes verified returning HTTP 200 with zero client console errors.

Latency & Guardrail Performance

- **Multimodal OCR to Journal Proposal:** Sub-3.2s end-to-end receipt parsing.
- **Deterministic Balance Gate:** Under 1.5ms mathematical validation (`Delta == 0`).
- **Global Edge First Contentful Paint:** Under 1.2s via Firebase CDN edge.
- **Zero Glassmorphism Guarantee:** High-contrast solid surfaces meeting WCAG AAA accessibility.

Future Development Roadmap

Phase 1: Foundation (Delivered)

- Autonomous SAK EMKM bookkeeping.
- PP 55/2022 statutory tax defense.
- 1,790-code KBLI 2020 verification.
- 24/7 background scheduled routines.
- Bilingual EN/ID user interface.

Phase 2: Open Banking (Q1 2027)

- Integration with Bank Indonesia Open Banking API (**BI-FAST**).
- Automated live bank feeds from BCA, Mandiri, and BRI.
- QRIS automated daily settlement matching.

Phase 3: Formal Credit (Q2 2027)

- One-click SAK EMKM audited financial pack submission to commercial banks.
- Automated credit underwriting scoring for government-subsidized **KUR** micro-loans.

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Official Submission Links

Live Deployed Applications

- **Production CDN Front Door:**
  https://gonata.web.app
- **Google Cloud Run Direct:**
  https://nata-web-615744448396.asia-southeast2.run.app

Source Code & Demonstration Video

- **GitHub Public Repository:**
  https://github.com/okihita/nata
- **Demonstration Video Walkthrough (Under 3 Minutes):**
  Synchronized Google Cloud Journey voiceover demo included in submission portal.

Team Zahrathustra — Google Cloud AI Builder Cup 2026

Convenience first, statutory compliance under the hood.